AWS Application Downtime Troubleshooting — Important Interview Questions & Answers

1.What would you check if an application is completely down?
First, I would check whether the EC2 instances are running and healthy.
Then I would check the ALB, target health, Security Groups, network connectivity, application service, and logs to find where the problem is.
Simple ga gurthupettuko:
EC2 → ALB → Target Health → SG → Network → Application → Logs.
Eng: I would check the EC2 instances, ALB, target health, network, application, and logs.

2.How would you check whether the problem is with the Load Balancer?
I would check the ALB status and target health.
If the targets are unhealthy, I would check the health check path, port, Security Group, and whether the application is running on the EC2 instances.
Simple ga gurthupettuko:
ALB problem ante ALB → Target Health → Health Check → Port → SG check chestham.
Eng: I would check the ALB, target health, health check, port, and Security Group.

3.What if the EC2 instance is running but the application is not working?
EC2 being in a running state does not always mean the application is working.
I would check whether the application service is running, whether it is listening on the correct port, and check the application logs for errors.
Simple ga gurthupettuko:
EC2 running unna application service run avthunda and correct port lo listen chestunda check chestham.
Eng: I would check the application service, listening port, and application logs.

4.What if ALB shows the EC2 instance as unhealthy?
I would first check the ALB health check configuration, such as the health check path and port.
Then I would check the EC2 Security Group, application status, and whether the health check request can reach the application.
Simple ga gurthupettuko:
Unhealthy Target → Health Check → Port → SG → Application check chestham.
Eng: I would check the health check, port, Security Group, and application.

5.What if the application works using the EC2 private IP but not through the ALB?
This usually means the issue may be with the ALB configuration, target group, health check, listener, or Security Group.
I would check whether the ALB listener is forwarding traffic to the correct target group and port.
Simple ga gurthupettuko:
EC2 direct ga work, ALB dwara work kaadu ante ALB → Listener → Target Group → SG check.
Eng: I would check the ALB listener, target group, port, and Security Groups.

6.What if the application suddenly stopped working?
I would check CloudWatch metrics and logs to understand when the problem started.
Then I would check recent deployments, EC2 health, ALB target health, application logs, database connectivity, and recent configuration changes.
Simple ga gurthupettuko:
Sudden issue ante Recent Changes → EC2 → ALB → Application → DB → Logs check chestham.
Eng: I would check recent changes, infrastructure health, application logs, and database connectivity.

7.How would you check whether the database is causing the downtime?
I would check whether the application can connect to the database and whether the database is available.
I would also check database connections, CPU, storage, Security Groups, and database logs for errors.
Simple ga gurthupettuko:
Application → RDS connection → SG → DB Health → Connections → Logs.
Eng: I would check database availability, connectivity, resources, Security Groups, and logs.

8.What if the application is working internally but users cannot access it?
I would check the public networking path, including Route 53, CloudFront if used, ALB, Internet Gateway, route tables, and Security Groups.
I would identify the exact point where the request is failing.
Simple ga gurthupettuko:
User → DNS → CloudFront → ALB → Network → EC2 path check chestham.
Eng: I would check the complete network path from the user to the application.

9.How would you troubleshoot an application after a recent deployment?
I would check whether the new application version was deployed correctly and whether the EC2 instances and ALB targets are healthy.
I would compare the new version with the previous working version and check application logs for errors.
Simple ga gurthupettuko:
New Deployment → Target Health → Application Logs → Previous Version compare.
Eng: I would check the deployment, target health, logs, and differences from the previous version.

10.What is your general approach to troubleshooting application downtime?
I would first identify whether the problem is at the DNS, Load Balancer, EC2, network, application, or database level.
I would check monitoring, logs, health checks, and recent changes to find the root cause and restore the application.
Simple ga gurthupettuko:
DNS → ALB → EC2 → Network → Application → DB → Logs.
Eng: I would identify the failed layer, check monitoring and logs, find the root cause, and restore the application.

11.What is the most important thing to remember for application downtime troubleshooting?
I should not randomly check everything. I should follow the request path step by step and identify where the request is failing.
Then I should fix the root cause and verify that the application is working again.
Simple ga gurthupettuko:
Request flow ni step-by-step follow chesi failure point identify cheyyali.
Eng: I would follow the request path step by step, identify the failure point, and fix the root cause.